Machine Learning-Operations-based Data Science Process Model – Material

This document contains all available materials of the MLOps-based Data Science Process Model (MLOps-DSPM). The detailed BPMN and UML information flow diagrams for each DS project's main stage are provided to enable a comprehensive understanding of the overall process.

This document covers all formalized diagrams that guide the execution of a DS project. The DSPM offers two types of diagrams. First, BPMN diagrams describe the DS workflow and the respective (mainly) responsible role categories of a DS phase or activity. The decision to structure the DS team roles as broader role categories, represented as BPMN swimlanes, is motivated by the fact that the title and precise configuration of a role largely depend on the project and organization. The five role categories are synthesized from the individual DS roles identified from the literature. The DS project stages and individual activities are mainly based on the literature review of DSPMs in Haertel et al. (2022). Secondly, UML information flow diagrams detail the information exchange and flow between information objects such as the MLOps components and documentation artifacts, as well as the DS role categories, depicted as actors.

The set of designed diagrams features a BPMN diagram and a UML information flow diagram for the entire DS lifecycle. Additionally, one BPMN diagram and one UML information flow are provided for each of the main stages of the DS lifecycle, with the intention of providing additional guidance for the project execution and setup. To ease the transition between (sub)models, color coding is included to clarify the assignment of tasks or components to the respective superordinate stage.

1 Overall Project View

The diagrams for the overall project view (Figure 1, Figure 2) aim to provide an abstract overview of the entire DS workflow, including the purpose and interaction of the MLOps components and the iterative nature of the DSPM. In the BPMN diagrams, each project phase/activity is assigned to a swimlane to indicate the main responsible role category for this task. This does not imply that other actors cannot be involved in this specific activity.

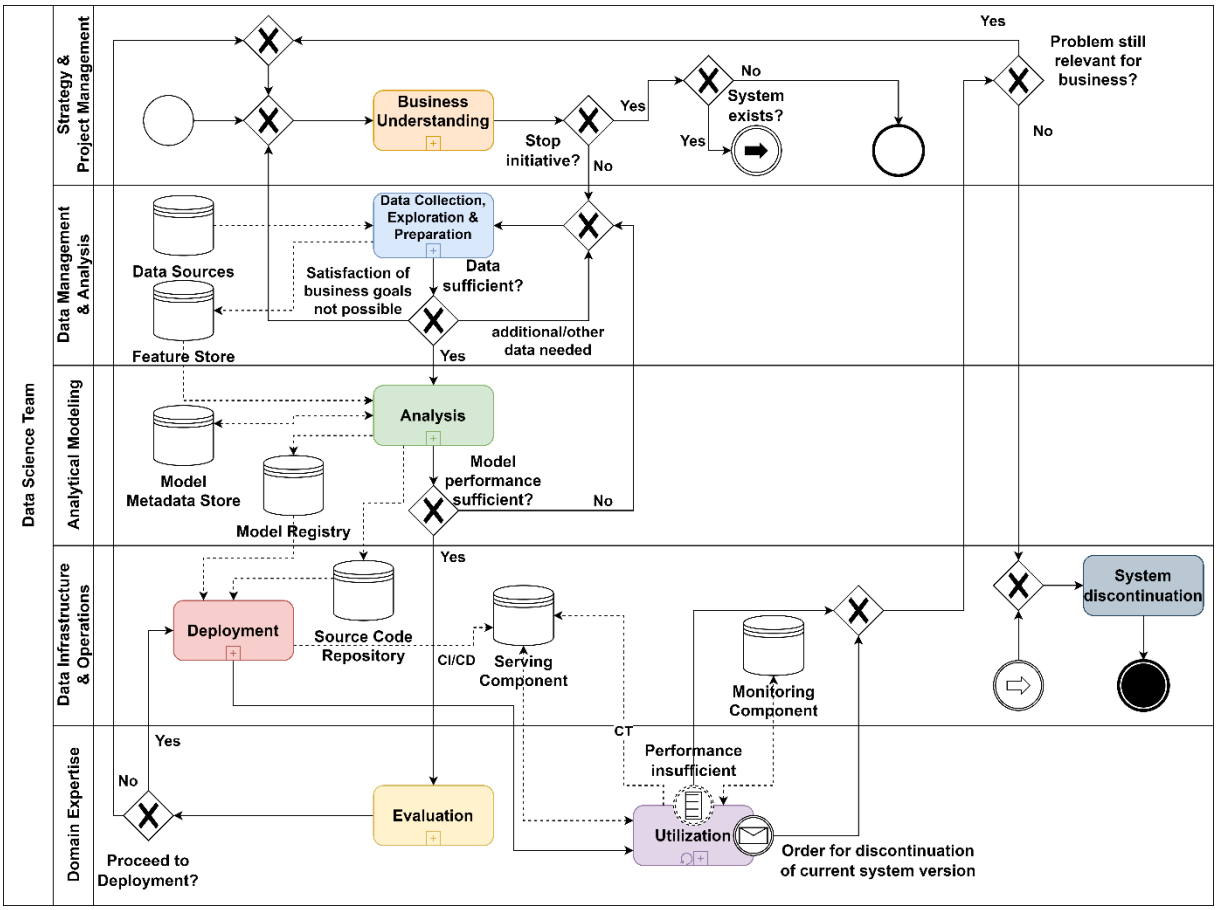


Figure 1. BPMN diagram for the overall project

Business Understanding constitutes the entry point for a DS project. The subactivities of this stage are detailed in Section 2. In case of deciding on a continuation of the initiative, the Data Collection, Exploration, and Preparation stage commences. Otherwise, the cycle ends after discontinuation of a potentially existing DS system from a previous iteration. The tasks of the second stage are detailed in the BPMN diagram in Section 3. Before continuing with the third phase, the deciding factor is the data. If additional data is needed, the stage is repeated to acquire further or other data. Additionally, a return to Business Understanding is possible.

The third stage, Analysis, is detailed in Section 4 and features the experimentation with various analytical models (e.g., machine learning models). MLOps components such as the feature store, model registry, and model metadata store come into play. In case of insufficient model performance, a return to the second stage is planned. Otherwise, the project continues with Evaluation, which is detailed in Section 5. Here, a decision regarding the continuation with the deployment is made, depending on the (potential) fulfillment of the business objectives with the current state of models. If this checkpoint decision is negative, a return to Business Understanding is prescribed.

Otherwise, the deployment process, detailed in Section 6, is initiated, which strongly depends on the form of the final product. A successful completion of deployment indicates that a model was made available to the domain users via the serving component. Utilization constitutes the last phase of the DS lifecycle, detailed in Section 7. The use of the deployed artifact in production for value creation is challenged by insufficient performance or the decision to discontinue the current system in place. Depending on the relevance of the problem that was addressed through the DS solution, a return to the Business Understanding is possible. Otherwise, the process ends.

The UML information flow (Figure 2) supplements this process view by clarifying the separation of environments and the information exchange between the (MLOps) components.

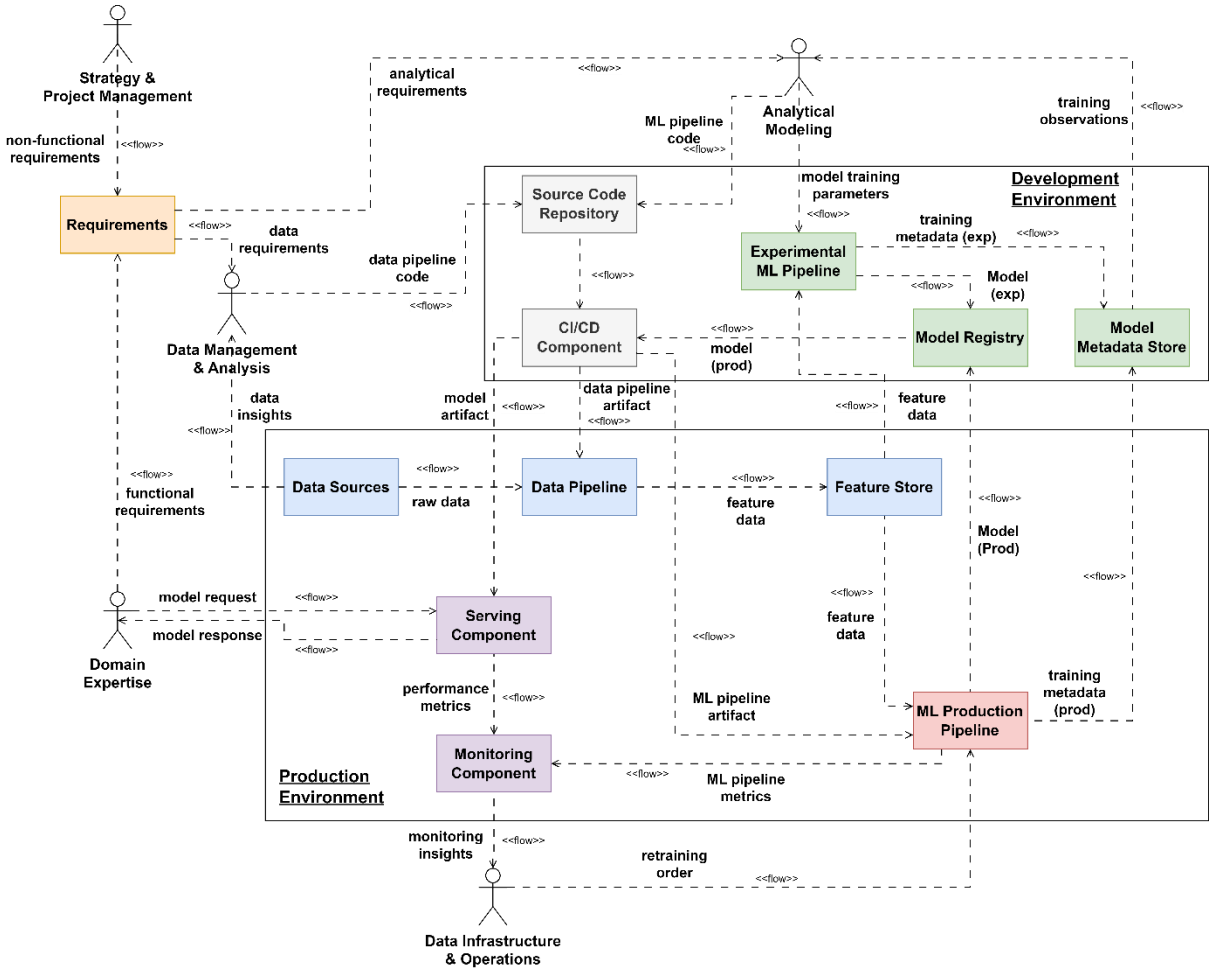


Figure 2. UML information flow diagram for the overall project

2 Business Understanding

The BPMN diagram (Figure 3) showcases the broader activities of the Business Understanding stage and the relevant documentation and project management artifacts. As the initial activity, Situation Assessment focuses on the analysis of the problem, including its suitability and the definition of requirements. Knowledge or best practices from previous initiatives can also serve as input. A concrete project plan is laid out upon the positive decision regarding the continuation of the undertaking. The output of this phase is the so-called DS project map.

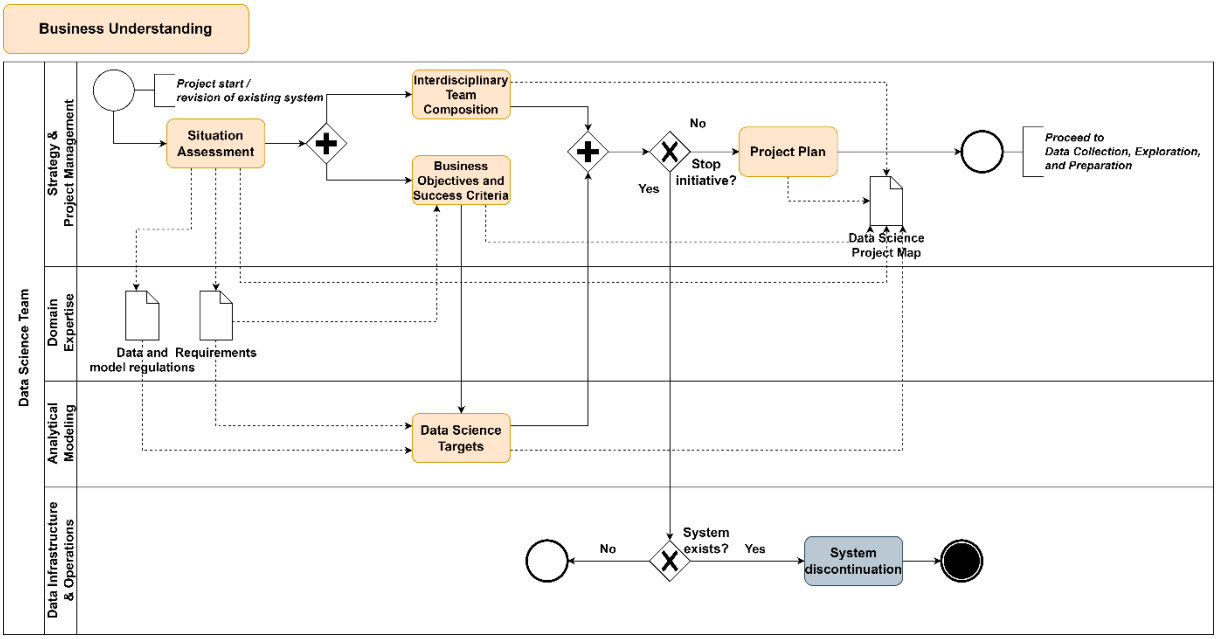


Figure 3. BPMN diagram for the Business Understanding stage

The corresponding UML information flow primarily details the input for the central artifacts of this stage.

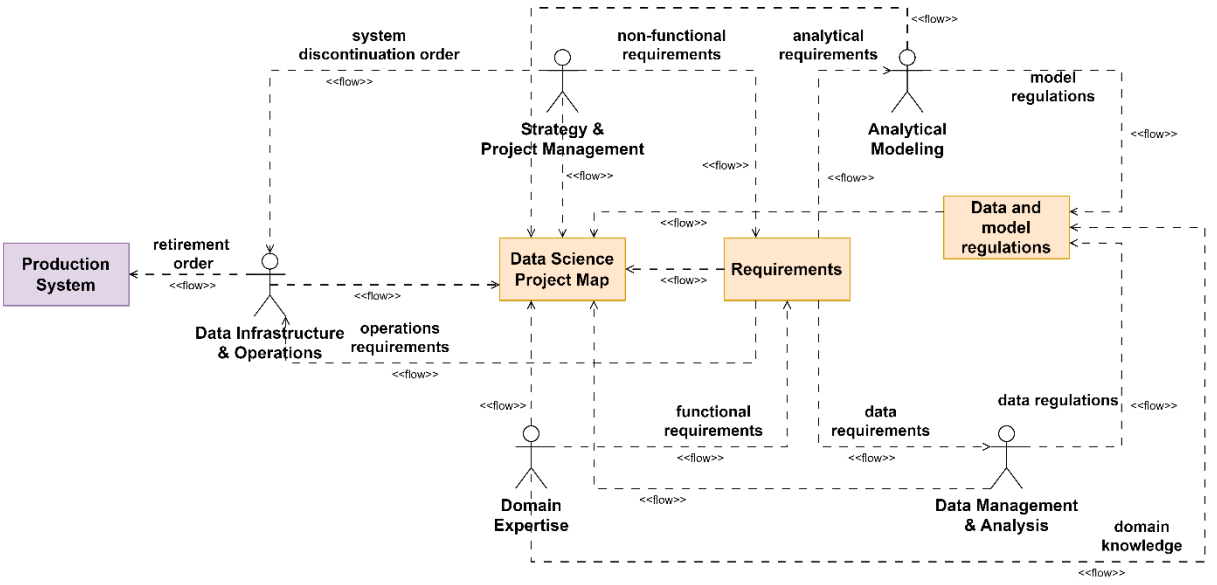


Figure 4. UML information flow diagram for the Business Understanding stage

3 Data Collection, Exploration, and Preparation

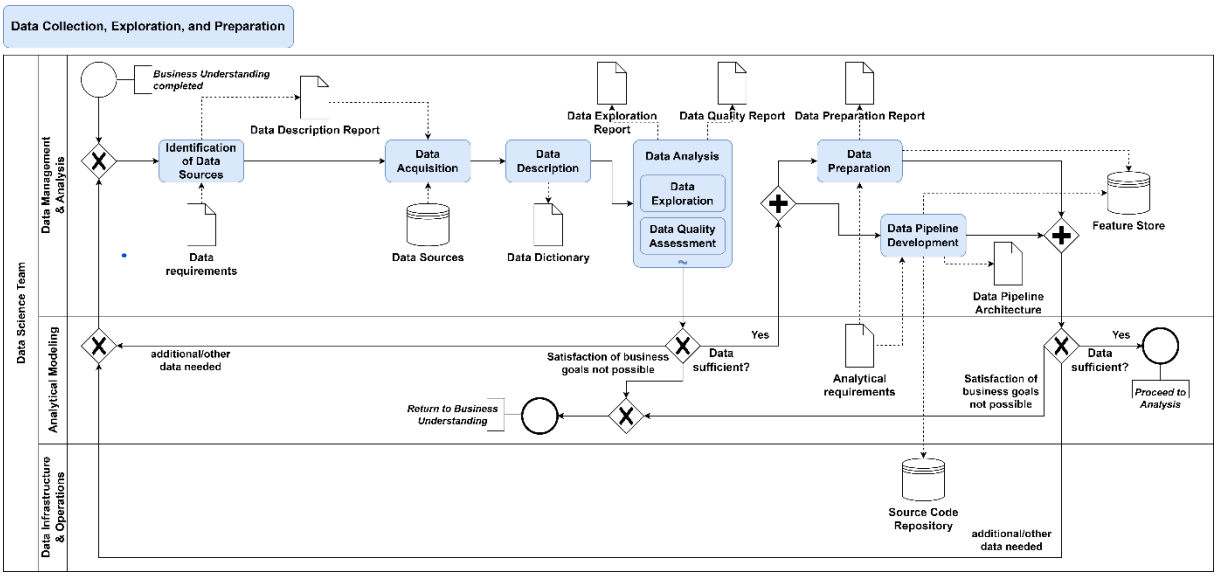


Figure 5. BPMN diagram for the Data Collection, Exploration, and Preparation stage

The BPMN diagram (Figure 5) showcases the broader activities of the Data Collection, Exploration, and Preparation stage, including various documentation artifacts for knowledge retention and important technical components. The corresponding UML information flow diagram (Figure 6) clarifies the interaction between these components further. After exploratory data analysis, the development of a data pipeline is a crucial task to automate the processing of relevant input data for the training of analytical models in the next stage. The data pipeline output is ingested into the feature store to allow for efficient and reusable feature retrieval.

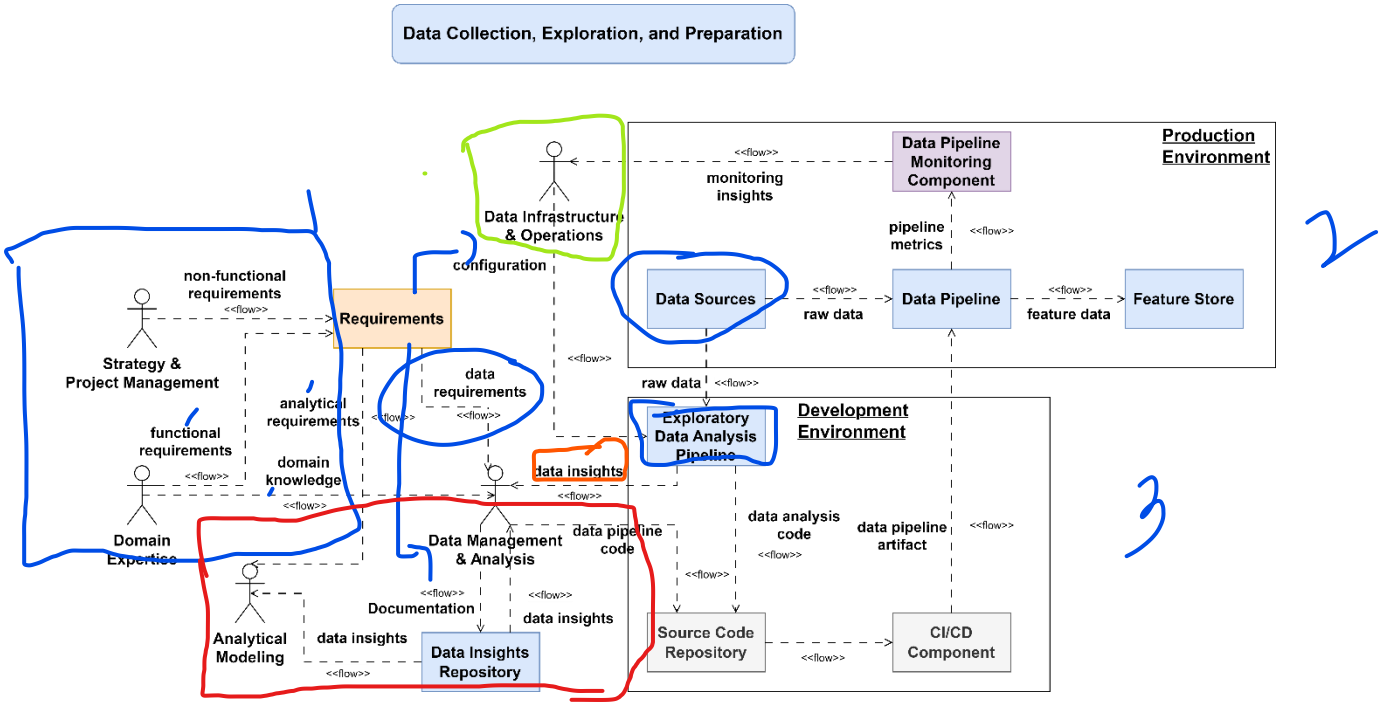


Figure 6. UML information flow diagram for the Data Collection, Exploration, and Preparation stage

The data pipeline code is held in a source code repository, and for its deployment to production, a CI/CD (Continuous Integration/Deployment) component is in place. The Data Insights Repository in the information flow diagram contains all relevant documentation artifacts for this phase.

4 Analysis

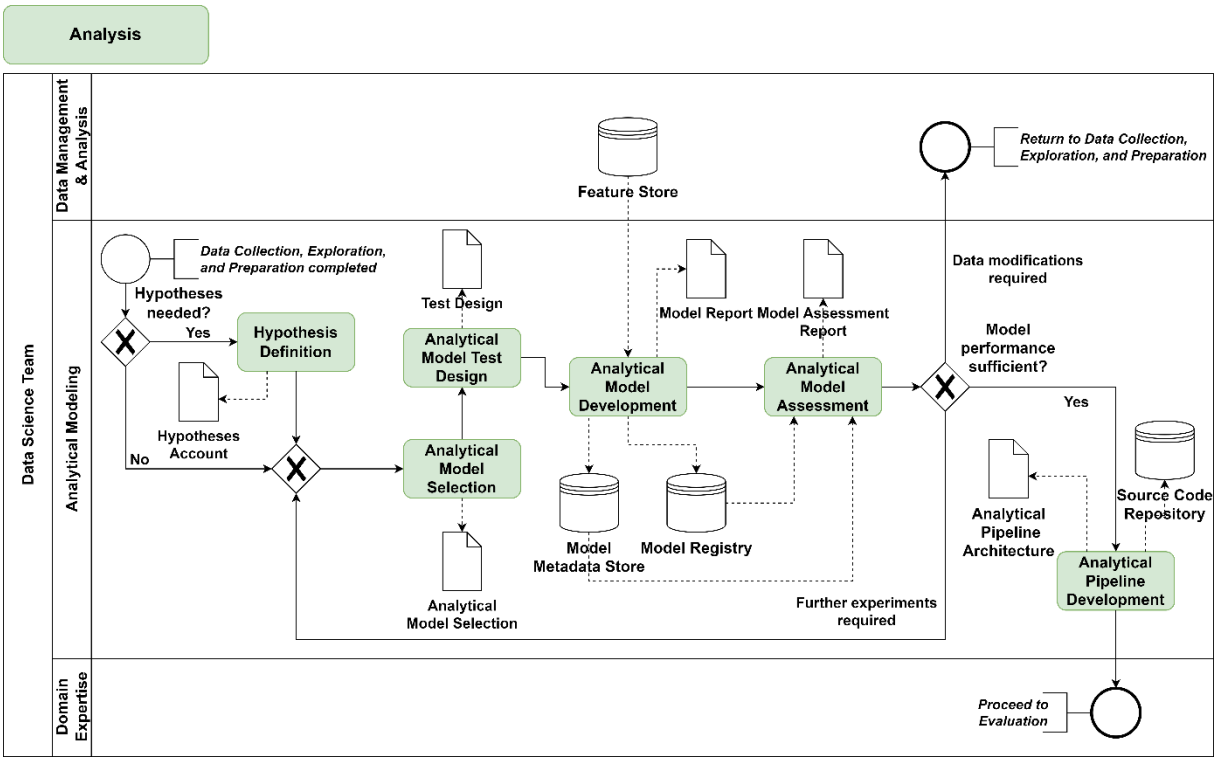


Figure 7. BPMN diagram for the Analysis stage

The BPMN diagram (Figure 7) showcases the broader activities of the Analysis stage, including various documentation artifacts for knowledge retention and important MLOps components to achieve transparency and reproducibility of the experiments. The corresponding UML information flow diagram (Figure 8) clarifies the interaction between these components further. The model metadata stores relevant metrics of training runs, while the model registry stores and versions the trained models. After achieving the envisioned model performance, the analytical pipeline development starts, as an ML production pipeline will be deployed later instead of just a model. The Modeling Insights Inventory in the information flow diagram contains all relevant documentation artifacts for this phase.

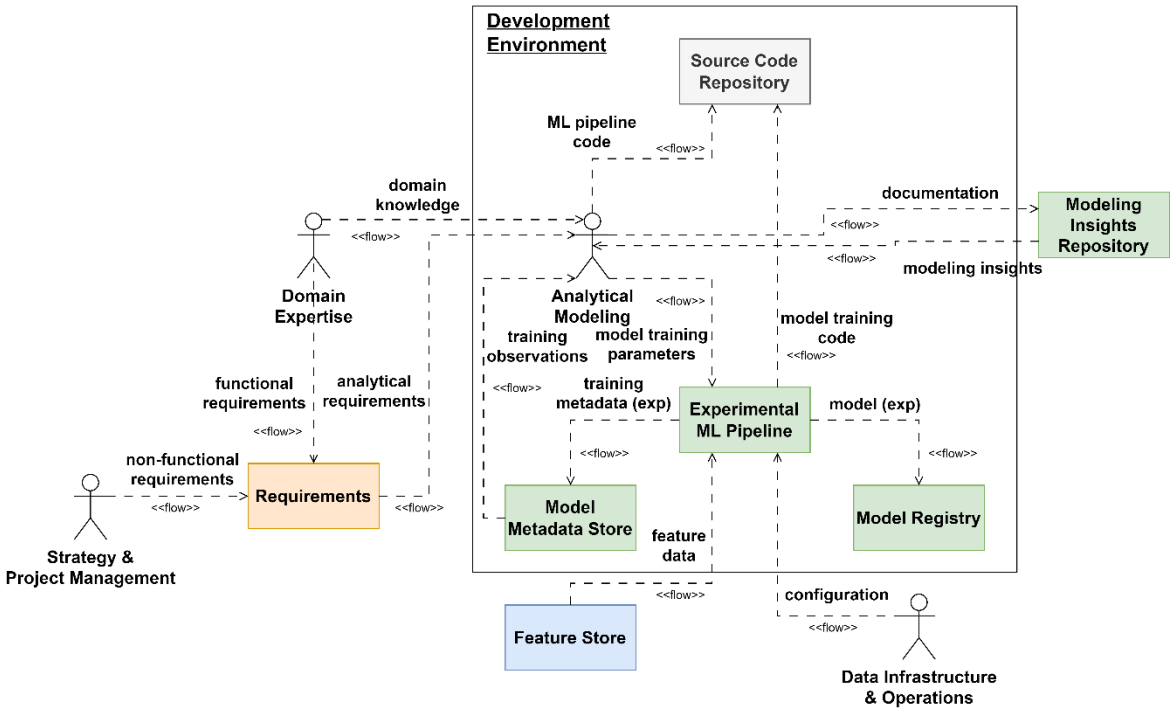


Figure 8. UML information flow diagram for the Analysis stage

5 Evaluation

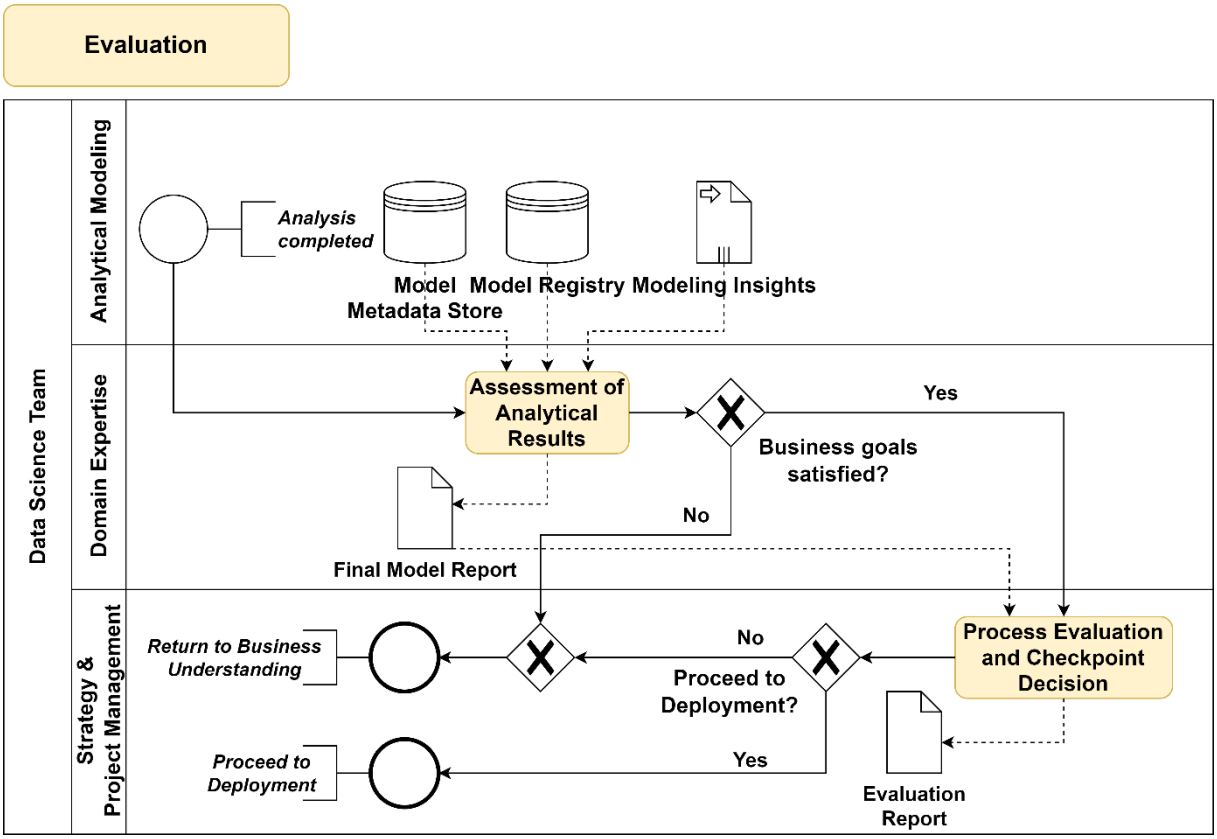


Figure 9. BPMN diagram for the Evaluation stage

The BPMN diagram (Figure 9) showcases the broader activities of the Evaluation stage, also including reports and the documentation artifacts from the previous stages that serve as important inputs for the assessment of the business objectives and the subsequent checkpoint decision. The UML

information flow diagram (Figure 10) further clarifies the exchange between relevant actors and information objects at this point.

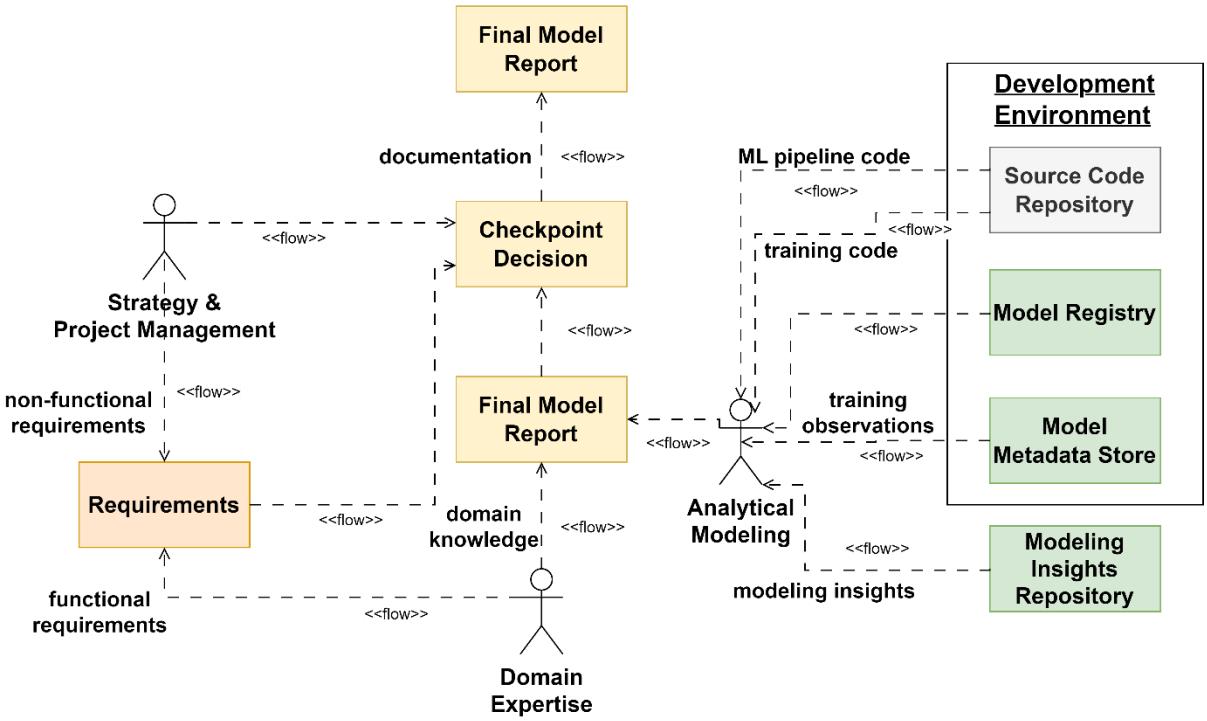


Figure 10. UML information flow diagram for the Evaluation stage

6 Deployment

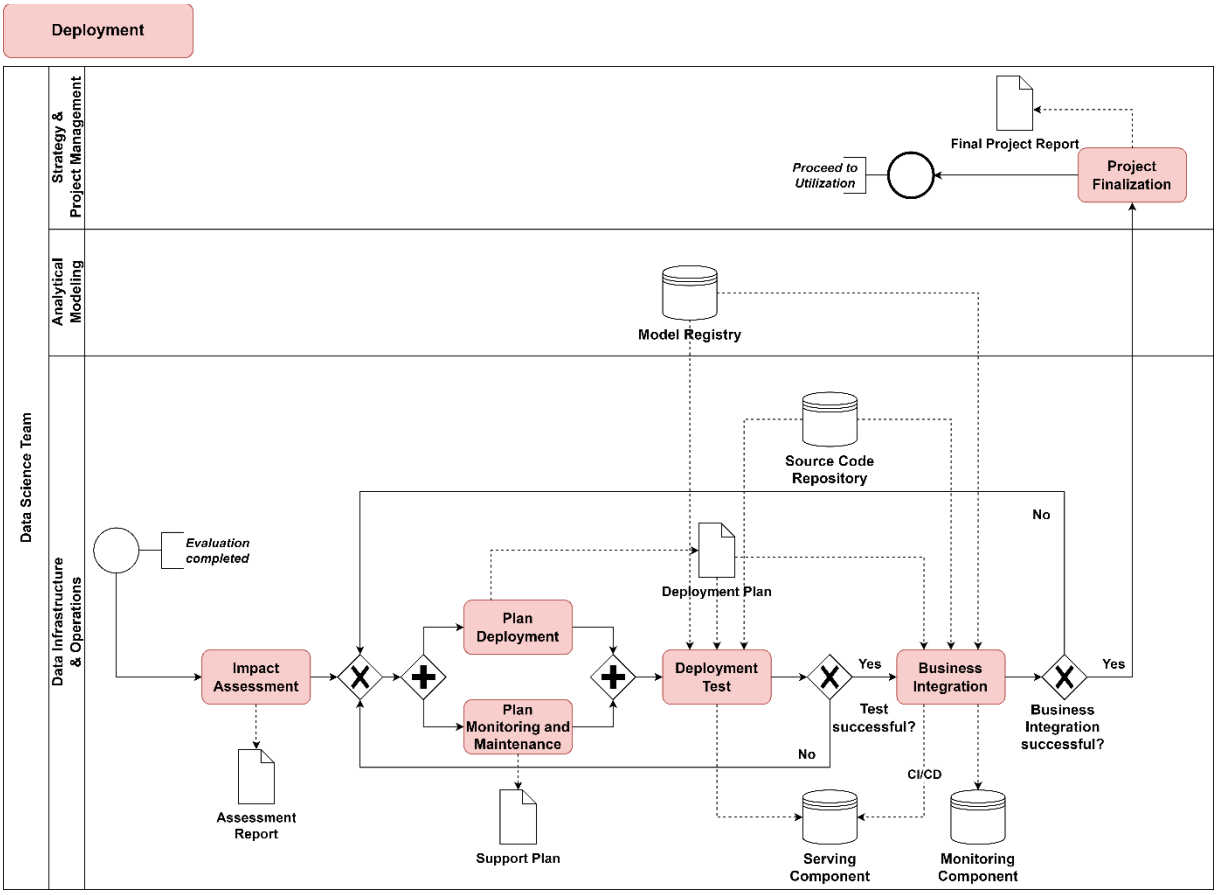


Figure 11. BPMN diagram for the Deployment stage

The BPMN diagram shows the broader activities of the Deployment stage, including important documentation artifacts and the purpose of the MLOps components. The initial task, Impact Assessment, evaluates the managerial and technical impact of the deployment of the analytical solution. Next to the deployment plan and test, planning the monitoring and maintenance of the DS solution to be deployed is an important consideration. Upon successful completion of the Business Integration, the DS project is finalized

As outlined in the UML information flow (Figure 12), the Business Integration does not merely involve the provisioning of a model to a serving component. Instead, as per the MLOps principles, an ML production pipeline is deployed. The execution of this pipeline, based on data provided by the feature store, trains a model that is then made available via the serving component. Similar to the Analysis stage, the model registry and model metadata store are also in place. The monitoring component observes both the pipeline and the deployed model.

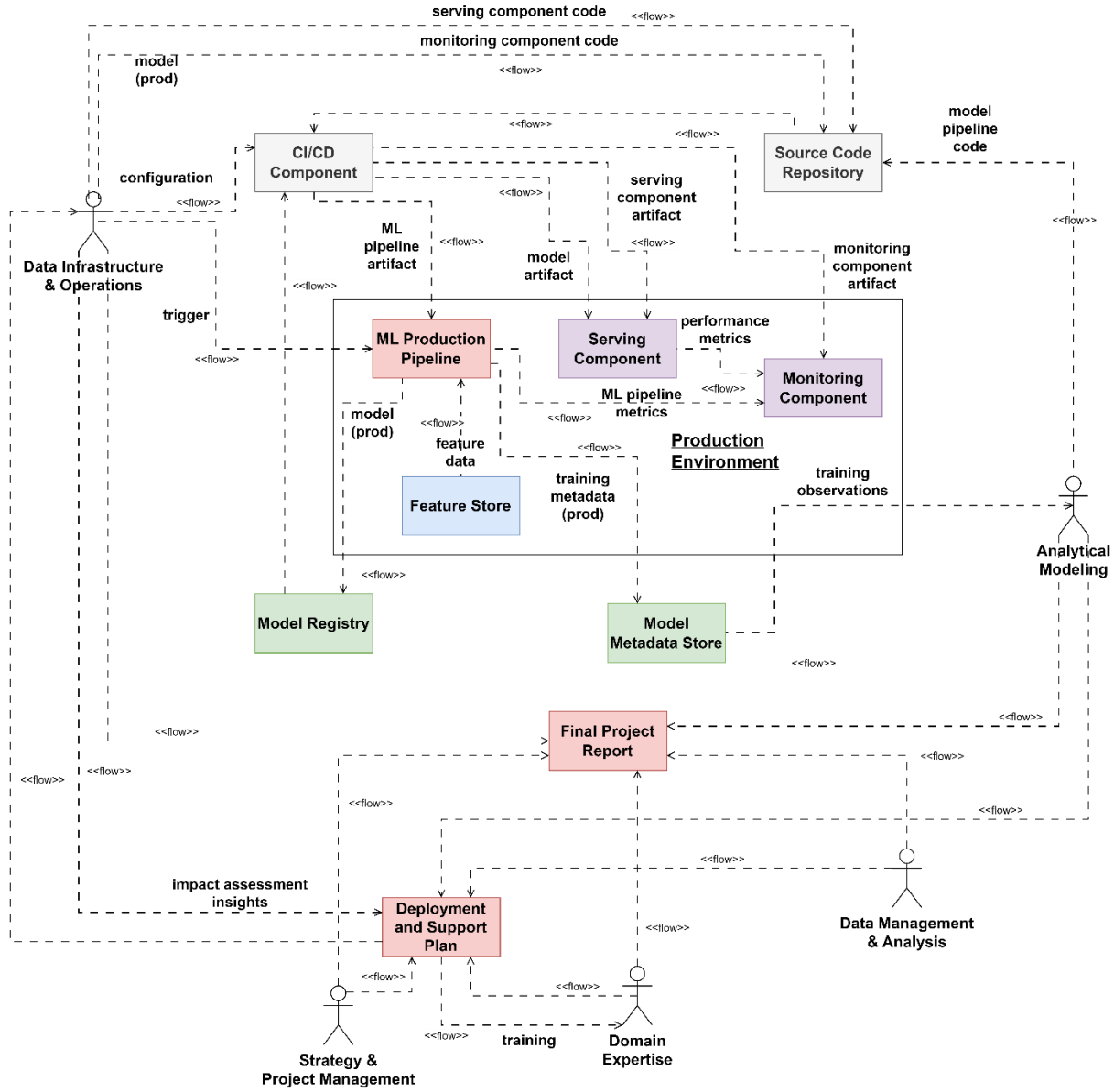


Figure 12. UML information flow diagram for the Deployment stage

7 Utilization

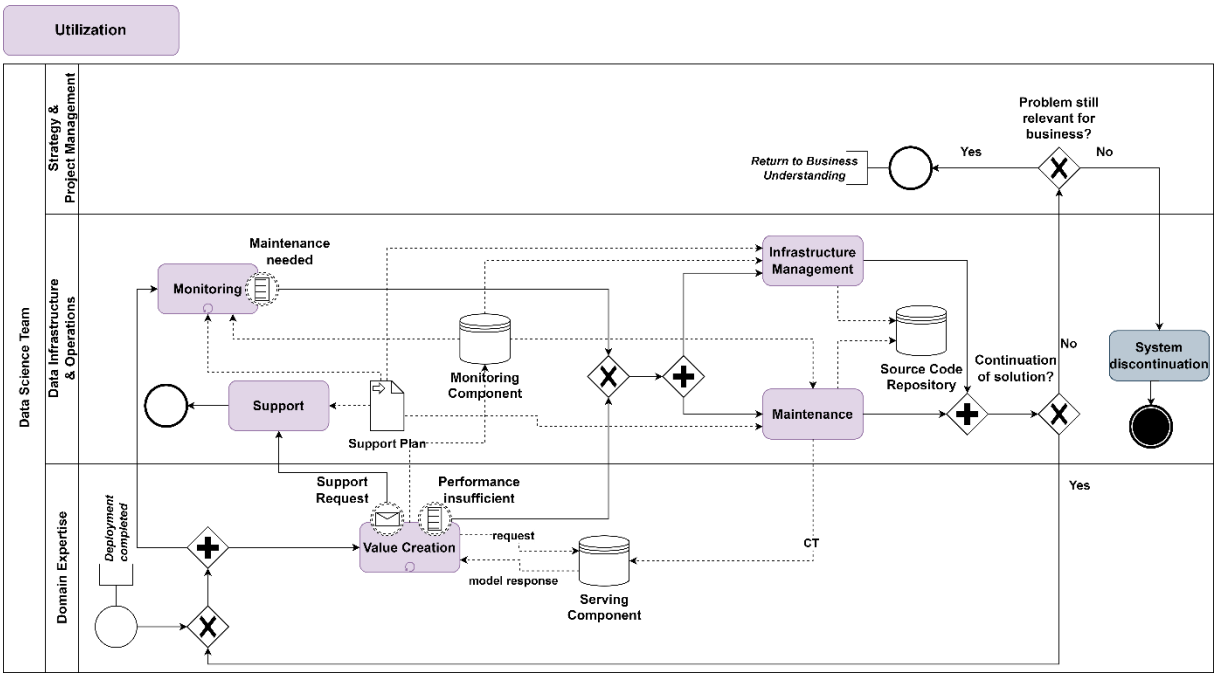


Figure 13. BPMN diagram for the Utilization stage

The BPMN diagram (Figure 13) shows the broader activities of the Utilization stage, including the main MLOps components. In essence, this stage comprises activities for the use of the DS system by domain users to achieve added value for the organization, monitoring and support, as well as maintenance tasks (e.g., model retraining) for continuous improvement (CT = Continuous Training).

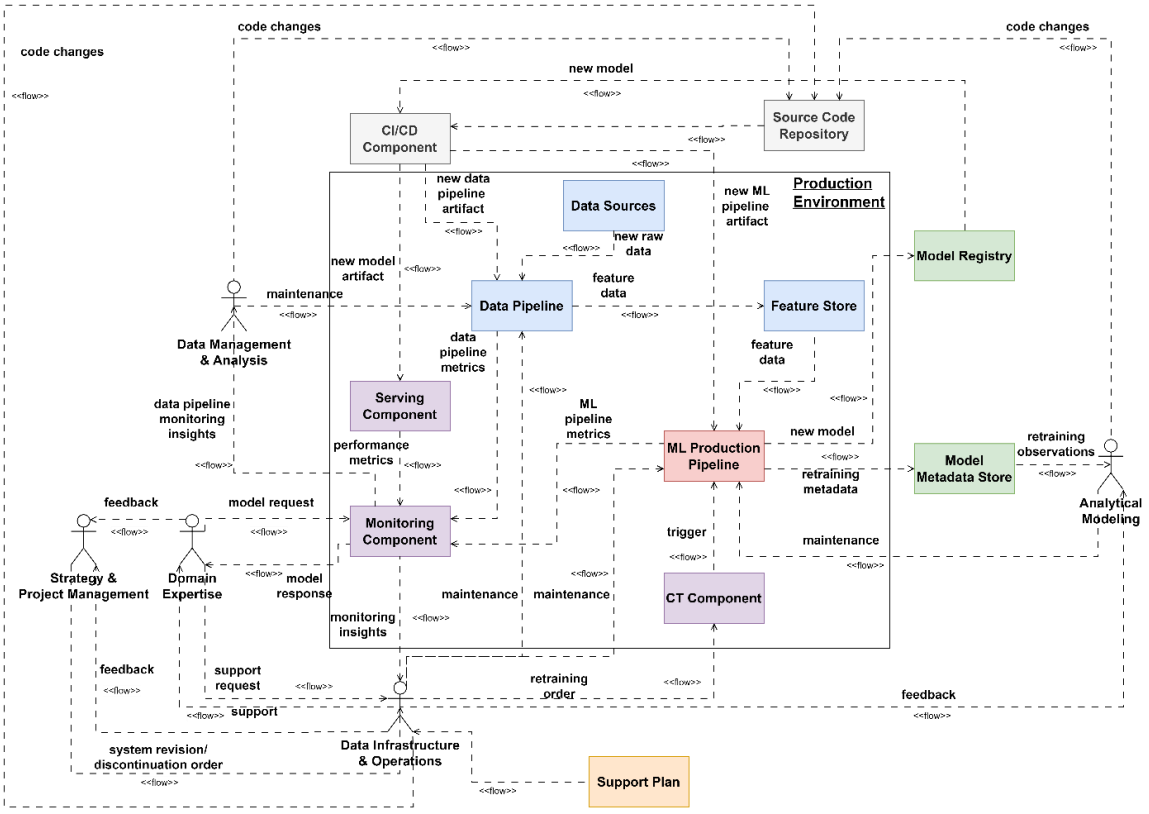


Figure 14. UML information flow diagram for the Utilization stage

The UML information flow diagram (Figure 14) for this stage further details the interaction between the components of the production environment to convey an understanding of how the Utilization activities are realized from an abstract technical perspective.

References

Haertel, C., Pohl, M., Nahhas, A., Staegemann, D., & Turowski, K. (2022). Toward A Lifecycle for Data Science: A Literature Review of Data Science Process Models. *PACIS 2022 Proceedings*.